

Ansible Ad-Hoc Commands

Quick, single-task CLI commands without playbooks

Definition

Ad-hoc commands are quick, one-off Ansible tasks run directly from the CLI **without a playbook**. They use Ansible modules to perform a single action on one or many managed hosts.

Basic Syntax

```
ansible <host-pattern> [-i inventory] -m <module> -a "<arguments>" [options]
```

- `host-pattern` – Target hosts, e.g. `all`, `web`, `db`
- `-i inventory` – Inventory file
- `-m module` – Ansible module to run
- `-a "args"` – Arguments passed to the module
- `-b` – Run with `become/sudo`
- `--check` – Dry run
- `--diff` – Show file changes

Common Examples

1. **Ping** – check connectivity and Python availability:

```
ansible all -m ping
```

2. **Shell** – run a shell command:

```
ansible web -m shell -a "uptime"
```

```
ansible all -m shell -a "df -h"
```

3. **Setup** – gather system facts:

```
ansible all -m setup
```

```
ansible db -m setup -a "filter=ansible.hostname"
```

4. **User** – create or manage a user:

```
ansible all -m user -a "name=deploy state=present" -b
```

5. **Other useful modules:**

```
ansible all -m copy -a "src=/tmp/app.conf dest=/etc/app.conf" -b
```

```
ansible web -m service -a "name=nginx state=restarted" -b
```

```
ansible all -m apt -a "name=curl state=latest" -b
```

When to Use

- Quick connectivity checks
- Troubleshooting
- Gathering facts
- One-time configuration changes
- Emergency restarts or checks
- Testing a module before writing a playbook

Limitations

- Not ideal for complex, repeatable, or multi-step automation
- No playbook structure, variables, roles, or handlers
- `shell/command` are not always idempotent
- Harder to version-control and audit than playbooks
- For repeatable work, use a **playbook**

Best Practices

- Prefer modules over `shell/command`
- Use `--check` first when changing systems
- Use `-b` only when needed
- Limit targets with host patterns
- Use `ansible-doc <module>` to see module options

Key Point

Ad-hoc = one task, fast, from CLI.

Playbook = many tasks, repeatable, versioned.